3D Path Planning of Unmanned Aerial Vehicle Based on Improved Grey Wolf Optimization Algorithm

1st Xin Liu
School of Automation Engineering
University of Electronic Science and Technology of China
Chengdu 611731, China
lx15713765446@163.com

2nd Yue Long*
School of Automation Engineering
University of Electronic Science and Technology of China
Chengdu 611731, China
longyue@uestc.edu.cn

3rd Tieshan Li
School of Automation Engineering Yangtze Delta Region Institute
University of Electronic Science and Technology of China
Huzhou, 313000, China
litieshan073@uestc.edu.cn

4th Tianpeng Huang
School of Automation Engineering
University of Electronic Science and Technology of China
Chengdu 611731, China
tphuang@uestc.edu.cn

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***Abstract*—Aiming at the problem of unmanned aerial vehicle's(UAV) path planning in complex battlefield environment, this paper proposes an improved grey wolf optimization algorithm (IGWO). UAV path planning is to plan the flight path from the starting point to the target point under the conditions of ensuring the best performance and satisfying different constraints. Based on the traditional GWO algorithm, this paper proposes two improvements: adjusting the decay form of the convergence factor, and re-updating the location according to weighted sum of ϕ, φ, χ. Furthermore, under the consideration of various constraints, a 3D path planning strategy for UAV based on IGWO is constructed to obtain the UAV path with the optimal comprehensive performance. Finally, the comparative simulations with three other algorithms are carried out. The simulation result shows that IGWO can be implemented for achieving optimal UAV path planning, and IGWO outperforms the other three optimization algorithms.**

***Index Terms*—Unmanned aerial vehicle; Path planning; Improved grey wolf optimization algorithm**

I. Introduction

The application of UAV in various fields continues to grow with improvement of control, communication and information technologies [1]. The range of applications in both military and civilian domains, such as, early warning, terrain detection, air strikes and target positioning, is expanding. Path planning is a premise to ensure the smooth execution of the UAV mission. It mainly refers to planning a flight path with as little cost as possible for a UAV from starting position to target position under the constraints of the flight performance and the complexity of the battlefield environment.

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Path planning plays an indispensable role in the stable operation of UAVs. In the past few years, swarm intelligent optimization algorithms have been extensively used for path planning of UAVs in a variety of environments. For example, in [2], a particle swarm optimization algorithm (PSO) is proposed and employed for real-time path of UAVs in a 3D environments. In [3], an ant colony optimization algorithm (ACO) is designed to create an obstacle free path for a UAV system. In [4], a Newtonian imperialist competitive algorithm (NICA) is proposed to create an optimal UAV flying path in three-dimensional space. In [5], an advanced swarm optimization method using bat algorithm (BA) is suggested for path planning of a UAV. In [6], an enhanced sparrow search algorithm called CASSA is put forward to tackle the path planning problem in three-dimensional space. In [7], an artificial fish swarm algorithm (AFSA) is developed and applied to global path planning. All the algorithms mentioned above can address the path planning problem of UAV. But how to effectively plan a optimal path in a specific environment using a suitable algorithm is the problem we need to consider.

In these path planning algorithms, Grey Wolf Optimization (GWO) [8] algorithm suggested by Mirtalili in 2014 is widely used in the design and optimization of UAV path. Compared with other planning algorithms, GWO has a obvious advantage of global convergence. In [9], GWO is is investigated to realize the trajectory planning of an autonomous submersible in a 2D underwater environment with static obstacles. In [10] and [11], GWO is introduced to global path design for mobile robots. In addition, GWO is also adopted for UAV path planning in [12] and [13]. However, it can be trapped in local optimum conveniently, so an IGWO is put forward in this paper. Based on the above description, the global searching abilities and speed of aggregation of the presented IGWO are improved by

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introducing the decay method of aggregation factor and the method of updating the position of weights.

The contributions of the paper are as follows:

1) The introduction of the nonlinear convergence factor a balances the iterative speed of the GWO algorithm, so that the late iteration speed is significantly improved.

2) The addition of the weight update method improves the accuracy of the fitness function value of the original algorithm. The method shortens the path of the UAV .

The structure of this paper is as follows. The modeling of the UAV path planning problem is introduced in the Section II. The principles of GWO and IGWO are described in detail in Section III. The specific steps of the IGWO algorithm for UAV path planning are introduced in detail in SectionIV. In Section V, the comparative simulations are performed to demonstrate the effectiveness of the developed approach and a summary discussion is offered. sectionVI concludes this work.

II. Analysis and Modeling of the UAV Trajectory Planning Problem

In this section, descriptions of tracking problem, terrain modeling and characteristics are presented.

A. Description of Track Problem

The description of the UAV's path planning problem can be expressed as follows. In the air combat environment under the constraints of given terrain obstacles [14] and no-fly zone threats, plan a flight path for the UAV from the starting point to the target point. The cost of the flight path of the drone is as small as possible. UAV needs to avoid obstacles during flight. Meanwhile, it cannot fly from no-fly zones to ensure their own safety. The costs include the UAV's own energy consumption, the length of the track, and the threat index of the track. Therefore, path planning is essentially an optimization problem.

The constraints of UAV's path planning can be divided into the following two aspects [15]: performance constraint and environment constraint. The first constraint includes: track segment length, flight distance, flight altitude, horizontal turn call, and climb/dive angle. The second constraint includes: enemy fire threat, radar monitoring and terrain obstacles.

B. Terrain Modeling and Characteristics

We assume that the constraint of the map is known. In the three-dimensional environment, how to obtain the ground height is one of the problems to be solved. A two dimensional matrix is created to display the ground. The value of every single member of the matrix is used to represent the height of the current position from the ground. When UAV performs various tasks in complex combat environments, it may face various threats of enemy firepower, such as anti-aircraft artillery and surface-to-air missiles. At present, radar is still the main equipment for long-distance detection and tracking of targets. The various threats of the enemy are simplified as mountain obstacles, and the parameters are changed to simulate the equivalent threat peaks. The constructed 3D simulation terrain is shown in Figure 1.

After the equivalent threat model is constructed, the simulation diagram can be obtained by setting the parameter values, where the starting point of the track is set as P_s (x_1, y_1, z_1) , and the target point is P_e (x_m, y_m, z_m). Divide the line between the starting point and the target point into m equal parts, The three-dimensional coordinates of the corresponding tracking point P_k are (x_k, y_k, z_k), which should meet the following two conditions:

1. The point complies with the boundary restrictions.

2. The point between P_k (x_k, y_k, z_k) and P_{k-1} $(x_{k-1}, y_{k-1}, z_{k-1})$ cannot exist in the obstacles.

The fitness function of the UAV can be expressed as (1)

$$C_t = C_h + C_l \tag{1}$$

where C_t is a total cost function, C_h is a threat index and C_l is the length of the generated path. The calculation formula of C_l and C_h can be given by

$$C_h = \sum_{k=1}^{m} \sqrt{(z_k - h_{best})^2} \tag{2}$$

$$C_l = \sum_{k=2}^{m} \sqrt{(x_k - x_{k-1})^2 + (y_k - y_{k-1})^2 + (z_k - z_{k-1})^2} \tag{3}$$

where z_k is the current flying height of UAV, and h_{best} is the best flying height of UAV. When UAV is flying at h_{best}, it is the safest and the threat index is the smallest. This situation does not appear in the actual environment, therefore, the calculation of the threat index also needs to be considered.

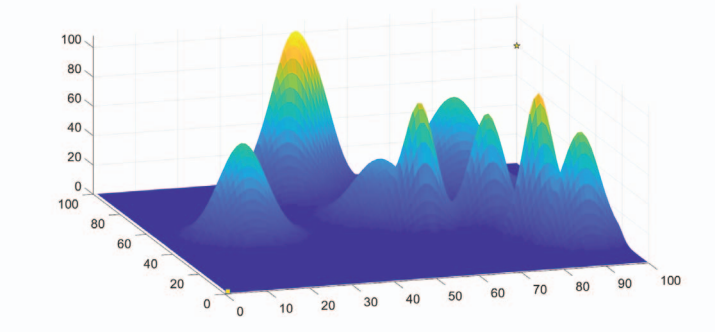


Fig. 1. 3D simulation map of threat terrain

III. GWO and its Improvements

A. Predatory Behavior of Grey Wolves

Grey wolf packs have very strict social hierarchies, and strict leadership levels prevailing. As shown in Figure 2, the wolf pack is a social organization with a very strict hierarchy. In a wolf pack, there are often four levels, namely wolf ϕ, wolf φ, wolf χ, and wolves ψ.

The first layer of the pyramid is wolf ϕ , the leader of the wolves. The hierarchical privilege of wolves is reflected in all

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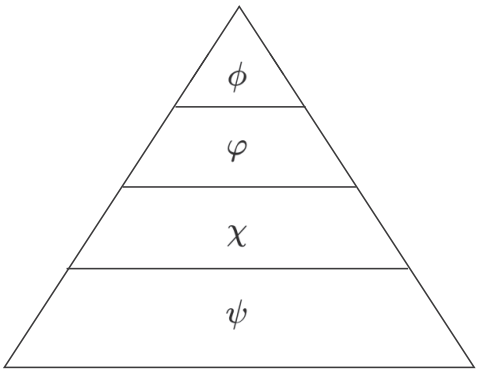


Fig. 2. Hierarchy of grey wolf

aspects. The wolf ϕ has the supreme right in the wolf pack and determines all the big and small affairs in the wolf pack, such as hunting, guarding, mating and so on.

The second layer of the pyramid is wolf φ , responsible for assisting wolf ϕ to make decisions. For example, when wolf ϕ in the wolf pack is killed by a hunter, or disappears for some reason, the second-ranked wolf in the pack will automatically fill in to become wolf ϕ.

The third layer of the pyramid is wolf χ , which mainly obeys the decision-making orders of wolf ϕ and wolf φ, and is primarily responsible for reconnaissance, sentry, guarding and other food. wolf ϕ and wolf φ with bad group fitness may also become wolf χ.

The bottom layer of the pyramid, i.e. wolves ψ, is primarily responsible for the balance of power within the population. It usually needs to obey wolves at other social levels. Although it seems that wolves ψ play little role in wolves, if there are no wolves ψ , the wolves will have internal problems such as cannibalism.

In addition, collective hunting is one of the social behaviors of grey wolves. A strict hierarchy runs through the hunting process of the entire group. Thanks to the strict hierarchy, the predation behavior of grey wolves is always successful.

B. Mathematical Model of Conventional GWO

In the design of GWO, the first step is to construct a model of the grey wolves pack. Then the fitness of each grey wolf in the pack was calculated, and the three grey wolves with the optimal fitness are labeled as wolf phi, wolf $varphi$ and wolf chi. That is to say, the order of social rank in the grey wolf group from high to low is: wolf ϕ, wolf φ, wolf χ and wolves ψ. The optimization process of GWO is guided by the best three solutions (referring to ϕ, wolf φ and wolf χ in each generation of population.

1) surround the prey: The behavior of the grey wolf to encircle and suppress the prey in the hunting process is defined as

$$\vec{E} = |\vec{B} \cdot \vec{X}_p(t) - \vec{X}(s)| \tag{4}$$

$$\vec{X}(t+1) = \vec{X}_p(t) - \vec{A} \cdot \vec{E} \tag{5}$$

note that (4) represents the distance between the prey and the grey wolf, and (5) is the update formula for the position of the grey wolf. In [4] and [5], t is the current iteration times. $\vec{E}$ represent the distance between abc and prey respectively. $\vec{X}_p$ is the position of the prey, and $\vec{X}$ is the position of the grey wolves pack. $\vec{A}$ and $\vec{B}$ are the coefficient vectors, given by

$$\vec{A} = 2\vec{a} \cdot \vec{r}_1 - \vec{a} \tag{6}$$

$$\vec{B} = 2 \cdot \vec{r}_2 \tag{7}$$

where $\vec{a}$ is the coefficient of convergence, it linearly reduces to 0 from 2 when the iterations increase., the modulo of $\vec{r}_1$ and $\vec{r}_2$ takes $\vec{a}$ random number between [0, 1].

2) hunt the prey: Grey wolf populations can identify and surround their prey. When the grey wolf recognizes the location of the prey, wolf α will lead other grey wolves to start surrounding the prey. In the decision space of many optimization problems, the optimal solution (i.e. the position of the grey wolf) is not known. We assume that the ϕ, wolf φ and wolf χ know the position of the prey best, save three optimal solutions to represent ϕ, wolf φ and wolf χ, and use their positions to predict the position of the prey, and the position of all wolves ψ are updated by the optimal grey individual wolf for position update. The mathematical model for individual grey wolves to track the position of their prey is as follows:

$$\begin{aligned} \vec{E_\phi} &= |\vec{B}_1 \cdot \vec{X}_\phi - \vec{X}| \\ \vec{E_\varphi} &= |\vec{B}_2 \cdot \vec{X}_\varphi - \vec{X}| \\ \vec{E_\chi} &= |\vec{B}_3 \cdot \vec{X}_\chi - \vec{X}| \end{aligned} \tag{8}$$

where $\vec{E}_\phi$,$\vec{E}_\varphi$,$\vec{E}_\chi$ represent the distance between abc and prey respectively, $\vec{B}_1$, $\vec{B}_2$, $\vec{B}_3$ are coefficient vectors, $\vec{X}_\phi$,$\vec{X}_\varphi$,$\vec{X}_\chi$ represent the current positions of ϕ, wolf φ and wolf χ, and $\vec{X}$ represents the current position of the grey wolves pack.

$$\begin{aligned} \vec{X}_1 &= \vec{X}_\phi - \vec{A}_1 \cdot (\vec{E}_\phi) \\ \vec{X}_2 &= \vec{X}_\varphi - \vec{A}_2 \cdot (\vec{E}_\varphi) \\ \vec{X}_3 &= \vec{X}_\chi - \vec{A}_3 \cdot (\vec{E}_\chi) \end{aligned} \tag{9}$$

where $\vec{X}_1$, $\vec{X}_2$, $\vec{X}_3$ represent the positions of the next ϕ, wolf φ and wolf χ respectively; $\vec{A}_1$, $\vec{A}_2$, $\vec{A}_3$ are the coefficient vectors. Obviously the next positions of ϕ, wolf φ and wolf χ can be obtained by (9). Furthermore, the final position of the wolves is defined as

$$\vec{X}(t+1) = \frac{\vec{X}_1 + \vec{X}_2 + \vec{X}_3}{3} \tag{10}$$

C. Defects of GWO

(1) The population diversity is bad. The initial population generated by random initialization cannot guarantee a good population diversity.

(2) Later iterations are slow. GWO mainly relies on the distance of the ϕ, wolf φ and wolf χ to judge the distance from the prey. Therefore, the convergence rate is slow in the later stage.

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(3) Fall into local optimum easily. Since wolf ϕ with the best fitness is not necessarily the global optimal solution, it is likely to be a local optimal solution, and the wolves ψ are infinitely close to the ϕ, wolf φ and wolf χ, therefore, it is likely to fall into the local optimal solution. This paper aims to address the problem of falling into local optimum.

D. IGWO Algorithm

From (8), it shows that the GWO algorithm updates the position to the individuals with high fitness value through trend search, which is the key group. Therefore, this will lead to weak global search ability, which may easily fall into local optima, especially when some high-dimensional datasets are used. So an IGWO, which uses the weighted sum of ϕ, wolf φ and wolf χ to re-update the position, is proposed. First, we change the convergence factor a uses an exponential function to decay. In (11), the value of u is [0, 1], and the value of p is [1, 2]. We can see that the convergence coefficient a reduces nonlinearly as the iterations increase. The calculation method of the weighted value can be obtained from (12). In (13), we replace the original position update method with the weight update position.

$$\vec{a} = 2x(1 - \frac{t^p}{T^p}) \tag{11}$$

$$\begin{aligned} u_1 &= \vec{A}_1 \cdot \vec{B}_1 \\ u_2 &= \vec{A}_2 \cdot \vec{B}_2 \\ u_3 &= \vec{A}_3 \cdot \vec{B}_3 \end{aligned} \tag{12}$$

$$\vec{X}(t+1) = \frac{u_1\vec{X}_1 + u_2\vec{X}_2 + u_3\vec{X}_3}{u_1 + u_2 + u_3} \tag{13}$$

IV. Steps and Design of Path Planning Based on IGWO Algorithm

Step 1: Parameter initialization. Set P_s and P_e of the UAV, initialize the individual, randomly select N tracks from P_s to P_e as the initial path, and initialize the position of the head wolf. Initialize the convergence factor $\vec{a}$, the coefficient $\vec{A}$ and $\vec{B}$, and the maximum number of iterations K.

Step 2: Calculation of the fitness function. If the current fitness function value is lower than the wolf ϕ fitness function value, update the wolf ϕ fitness function value. If the current fitness function value is higher than wolf ϕ but lower than wolf φ, update the wolf φ fitness function value. If the current fitness function value is lower than wolf χ but higher than the wolf φ fitness function value, update the fitness function value of wolf χ.

Step 3: Update of individual parameters. Start to update the positions of the ϕ, wolf φ and wolf χ through for (8) and (9), and update the respective $\vec{A}$ and $\vec{C}$ of the three wolves at the same time.

Step 4: Update of individual fitness. Calculate the fitness function value of all grey wolves, and judge whether they meet the boundary constraints and whether there is a collision. After these two conditions are met, update the fitness of ϕ, wolf φ and wolf χ. The weight update method is used to update the position of grey wolf pack, and after K is reached, the track is output. The flow chart is shown in Figure 3.

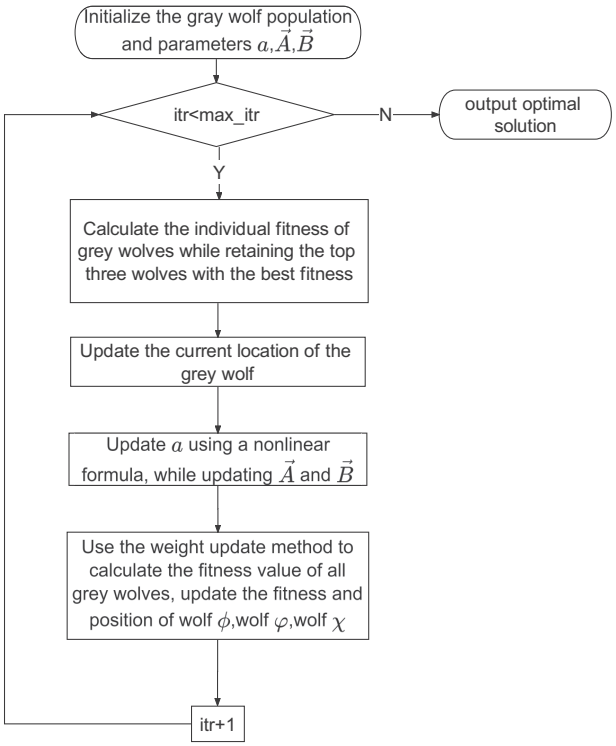


Fig. 3. UAV track planning flowchart with improved gray wolf optimization algorithm

V. Algorithm verification and simulation

In simulation, the UAV flight space is a three-dimensional space of 120km*120km*120km. P_s is (1, 1, 1) km and P_e is (120, 120, 60) km. IGWO is used to compare with the three algorithms : PSO, GWO and AFSA. Since the algorithm is accurate to 3 decimal places when the algorithm is iterated for 40 times, the maximum number of iterations is set to 60. In order to facilitate the comparison of these four algorithms, each map is randomly generated.

Figure 4 and Figure 5 show the main picture and the overhead picture of optimal flight paths planned by the four algorithms in a simple environment. One can get that each algorithm can efficiently avoid threats and do not exceed the map range.But the track path generated by IGWO is shorter and smoother.

Figure 6 shows the relationship between the iteration times of the four algorithms and their optimal fitness function values. From the result, we can see that IGWO reaches its own optimal fitness function value with less iterations than the other three algorithms. Its optimal fitness function value is the lowest, one can get that IGWO is obviously outperformed by three other algorithms..

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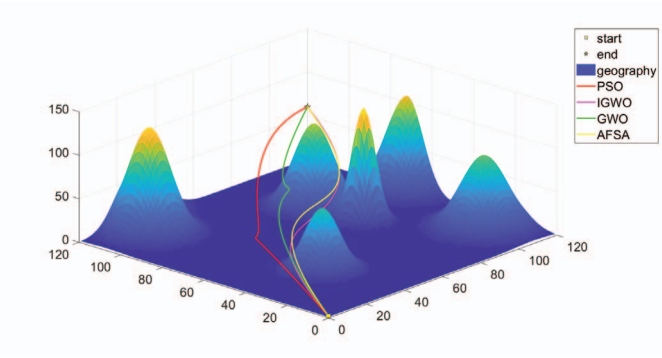


Fig. 4. Main picture of the optimal flight paths plotted by the four types of algorithms(under a simple environment)

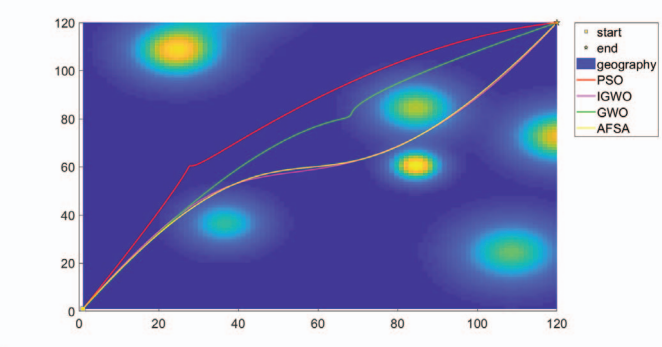


Fig. 5. Overhead picture of the optimal flight paths plotted by the four types of algorithms(under a simple environment)

Next, the UAV's path planning under the complex environment is considered. The results of optimal flight paths' main picture and overhead picture by the four algorithms are plotted in Figure 7 and Figure 8 respectively. One can get that in Figure 7 and Figure 8 that although AFSA has effectively avoided obstacles at this time, it has exceeded the scope of the map. At this time, the other three generated tracks effectively avoid obstacles and do not exceed the map range. Meanwhile, one can get that in Figure 7 and Figure 8 that the track path generated by IGWO is shorter and smoother.

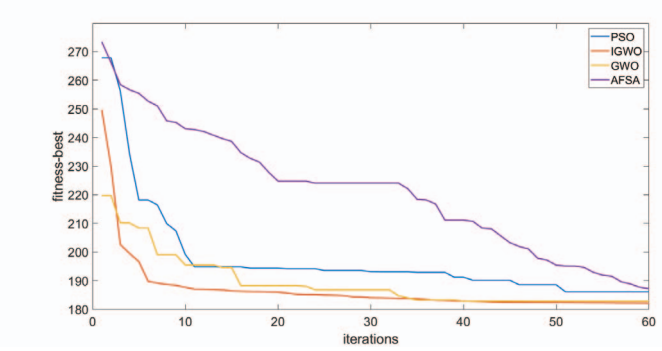


Fig. 6. The optimal fitness function values of the four algorithms(under a simple environment)

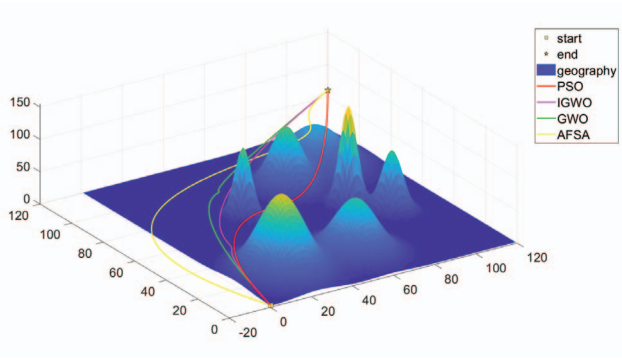


Fig. 7. Main picture of the optimal flight paths plotted by the four types of algorithms(under a complex environment)

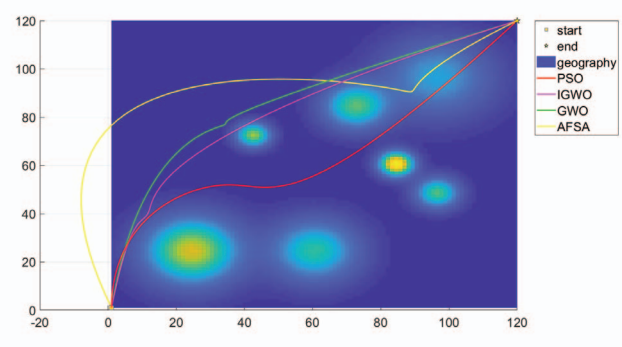


Fig. 8. Overhead picture of the optimal flight paths plotted by the four types of algorithms(under a complex environment)

Figure 9 displays the relations as between the value of the optimal fitness function and the number of iterations in a complex environment. We can see that IGWO can still achieve the optimal fitness function value with the least number of iterations under the complex environment. And its optimal fitness function value is the lowest, therefore, we obtain that IGWO does outperform the other three algorithms.

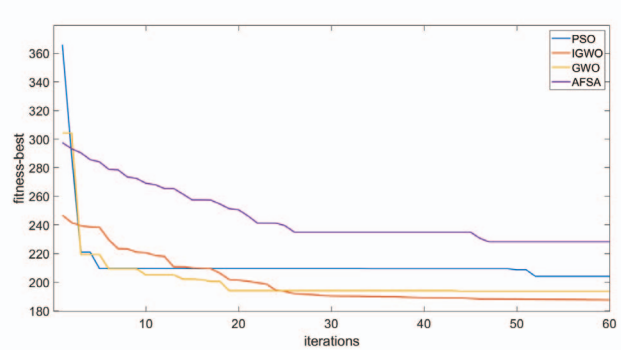


Fig. 9. The optimal fitness function values of the four algorithms(under a complex environment)

Through the comparison of the UAV's path planning under two different kinds of environments, the path planning based on IGWO is superior to the other three algorithms in terms

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of the number of iterations and the final optimal fitness function value. The proposed algorithm has certain practical significance.

Table 1 demonstrates the compared performance of these four types of algorithms. Since the data generated by the two different kinds of environments point to the same conclusion, this paper only releases the performance comparison data of the first group. In Table 1, the track length using IGWO is the shortest, and the curve is relatively smooth. One can see that the other three algorithms are all trapped in a local optimum, and the iterative average time of IGWO is shorter.The simulation outcomes demonstrate that the IGWO algorithm can be an ideal solution to the UAV path planning problem.

TABLE I
PERFORMANCE COMPARISON OF FOUR ALGORITHMS

Algorithms Comparison	**Performance Index**	
	Track Length	***Iteration Average Time***
GWO	182.3575	1.2559
IGWO	180.3336	0.9599
PSO	190.8951	1.1134
AFSA	191.5269	1.3556

VI. CONCLUSION

This paper aims at addressing the proplem of path planning of UAV under in complex battlefield envirment. When the three-dimensional map, starting point, and ending point were known, the proposed IGWO algorithm was employeed for UAV path planning, and the three algorithms of PSO, GWO, and AFSA were used for comparison. The simulation outcomes indicate that the path planning of the UAV using these four algorithms can make it feasible to avoid obstacles. However, the IGWO algorithm yielded shorter path lengths and comparatively smoother paths. In addition, as the iterations are 20, the IGWO algorithm has reached the optimal fitness, indicating that a smaller number of iterations can already achieve better results.

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